

2017 H2 Q4

Section: Our Dynamic Universe

Topic: Waves analogy & traffic flow ($v = f\lambda$)

Motorways with variable speed limits use overhead systems to choose the displayed maximum speed. They observe the traffic 'flow' and use the relation speed = frequency $\times$ wavelength ($v = f\lambda$). Use physics to comment on this system.

Worked commentary

- Correct physical relation: For a steady stream of equally spaced cars, frequency f (cars per second passing a point) times spacing λ (m per car) gives average speed $v = f\lambda$. This is directly analogous to waves ($v = f\lambda$) and to traffic-flow theory ($v = q/k$).
- When it works best: The model assumes near-uniform spacing and speed (steady flow). Under these conditions the displayed limit based on $v = f\lambda$ should match the actual safe mean speed and can smooth traffic.
- Benefits of lower limits in heavy traffic: Reducing the displayed speed before density becomes too high can prevent stop-start 'shock waves' that propagate backwards through queues. Smoother, slightly slower traffic can increase overall throughput and reduce collisions.

- Limitations/uncertainties: Real traffic has variable spacing and driver reaction times; merging, incidents, and lane changes break the uniform-flow assumption. If f or λ are measured over too short a time or too small a sample, the estimate of v can be noisy or lag actual conditions.
- Practical conclusion: The $v = f\lambda$ approach is physically sound as a control metric, provided measurements are averaged over suitable times/distances and safety margins are applied. Used proactively, variable limits can stabilise flow and reduce congestion.

Revision tips

- Wave relation $v = f\lambda$ applies to streams (cars per second $\times$ spacing per car).
- Traffic 'shock waves' move backwards through lines of cars when density gets too high.
- Smoothing speed variations often improves flow and safety more than raising the limit.